

# Robust Pricing in Marketing Channels

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**Abstract.** In many cases, managers cannot rely on expected profit maximization to determine prices because of a lack of reliable sales data. To overcome this issue, the pricing literature suggests adopting strategies that are robust to model uncertainty. However, most of this literature addresses situations where firms sell directly to consumers, leaving a gap in understanding how to implement robust pricing strategies when products are sold through retailers. To address this gap, we propose a model that explores the impact of robustness on pricing strategies for both manufacturers and retailers. The model provides three novel insights. First, we find that in equilibrium, channel partners will seek different levels of robustness for their pricing decisions. Second, the model reveals that injecting robustness in channel pricing decisions can not only help manage unforeseeable demand shocks, as it should be, but also mitigate double marginalization. Finally, the model uncovers that a decentralized channel can generate more profit than the centralized channel when robust pricing strategies are implemented.

**Keywords:** marketing channels • robust pricing

## 1. Introduction

Profits go down when incorrect demand predictions inform pricing decisions. In the case of new products for example, prelaunch demand estimation often yields inaccurate sales forecasts that lead companies to pricing mistakes; for example, in 2007 Apple significantly slashed the price of its first iPhone just two months after the initial launch to better meet demand (Wingfield 2007), thus impacting not only itself but also the companies distributing the new technology. Global brands often fail their entries in foreign markets because of erroneous evaluations of local market conditions (Sainam and Bahadir 2024), not unlike what happened to the U.S. company Kellogg when it attempted to enter the Indian market for convenience breakfast cereals (Dawar and Chattopadhyay 2002). In the same vein, uncertain consumers' acceptance of new channels (Geyskens et al. 2002) can also lead to inaccurate demand estimation and therefore wrong prices, impacting both brands and retailers.

Inaccurate demand assessments occur for various reasons, for example, absence or scarcity of actual market data (see, e.g., Cohen et al. 2022), unreliable data collection (see, e.g., Sridhar et al. 2017), instability of market conditions (see, e.g., Bronnenberg and Mahajan 2001, Rubel et al. 2011, and Adulyasak et al. 2023), or mis-specified models (Rubel and Naik 2017). As a result, uncertainty exists about what the true demand model is, that is, uncertainty about sales' true probability distribution. Why? Because the lack of reliable data or the use of an incorrect estimation procedure prevent

discovering the statistical properties of demand via the estimation of a sales response model. Therefore, following the traditional approach of expected profit maximization to choose prices under uncertain demand can be perilous because an ambiguity exists about sales' true probability distribution.

To address this challenge, a rich literature on robust pricing exists to guide managers' pricing decisions in such situations where sales data are nonexistent, unreliable, or scarce, that is, pricing decisions under uncertainty when probabilities themselves are unknown. This literature departs from the Bayesian paradigm and injects robustness in pricing decisions by adopting a minimax approach; that is, Bergemann and Schlag (2008) and Caldentey et al. (2017) applied the minimax regret criterion, and Besbes and Zeevi (2011) and Adida and Perakis (2010) applied robust optimization, whereas Lim and Shanthikumar (2007) used robust control. Even though the extant robust pricing literature has covered a rich set of managerial contexts such as monopoly pricing (Bergemann and Schlag 2011, Cohen et al. 2021), competitive pricing (Perakis and Sood 2006), new product pricing (Handel and Misra 2015), markdown pricing (Mersereau and Zhang 2012, Dziecichowicz et al. 2015), and even revenue management (Lan et al. 2008, Perakis and Roels 2010), it has focused exclusively on products sold directly to consumers and has not taken into consideration alternative channel structures.

Firms, however, often sell products through indirect marketing channels (Ingene and Parry 2005, Ingene

et al. 2012). Therefore, when managers implement pricing strategies based on incorrect demand estimates, they affect not only their own profits but also the profits of their channel partners. In the consumer electronics industry, for example, firms frequently launch new devices without reliable data on consumer adoption rates. Manufacturers like Apple or Samsung often propose suggested retail prices and wholesale terms before market response is known, whereas retailers must decide whether to follow these pricing recommendations or adjust them to better reflect local market conditions. In the personal care and beauty sector, manufacturers may set pricing guidance for new product lines, but retailers, especially in international markets, must interpret and apply these guidelines in environments where consumer preferences are highly uncertain. Likewise, in agriculture, upstream input providers (e.g., seed or fertilizer manufacturers) and downstream distributors must make pricing decisions long before the actual demand, driven by a litany of factors such as weather, crop cycles, or global commodity prices, can be observed. In all of these channel settings, pricing decisions are made under ambiguous demand conditions, with both manufacturers and channel partners jointly exposed to financial risk.

Therefore, implementing robust pricing strategies must take into consideration the channel strategy used to sell products, for example, selling directly versus indirectly to consumers, the resulting strategic interactions between channel partners, and their respective needs for robustness. Specifically, (i) how should robust pricing strategies differ from non-robust pricing strategies in marketing channels, and (ii) how does the pursuit of robustness impact channel efficiency and the need for channel coordination?

To address these questions left unanswered by both the robust pricing literature and the marketing channel literature, we propose a Stackelberg game between a manufacturer and a retailer, where the manufacturer acts as the leader and the retailer as the follower. The uncertain demand shocks affecting demand make the game stochastic. However, these shocks are drawn from a probability distribution that is unknown to both players at the time of making pricing decisions. To solve the pricing game given demand's unknown probability distribution, we adopt a minimax control approach (see, e.g., Bernhard 2002, Basar and Bernhard 2008, and Hansen and Sargent 2008), which allows us to characterize the optimal robust pricing strategies for both the centralized and decentralized channel structures and to understand the value of robust pricing strategies in marketing channels.

We first show that levels of robustness are inversely proportional to the average unit margins each channel partner anticipates capturing based on demand's nominal model. As a result, channel partners do not

necessarily choose similar levels of robustness. We then show that the value of robustness in marketing channels is twofold; that is, it can not only help manage unforeseeable adversarial demand shocks but also, under certain circumstances, improve channel efficiency by mitigating double marginalization. Finally, we find that the robust decentralized channel can generate more profit than the robust centralized channel, which invites questioning the value of channel coordination mechanisms under model uncertainty. These results bridge the gap between the robust pricing literature and the marketing channel literature.

In the next section, we introduce the concept of robustness. In Section 3, we characterize the optimal robust pricing strategy in a vertically integrated channel and document when it creates value for the firm. In Section 4, we derive the optimal robust pricing strategies for channel partners in a decentralized channel and show when these pricing strategies create value for each player. Finally, we conclude the paper in Section 5.

## 2. Robustness

The need for robustness comes from the risk of making decisions based on an incorrectly specified sales response model. Following the literature on robustness (see, e.g., Hansen and Sargent 2008), we conceptualize a model as being a probability density function that allows the decision-maker to assign a probability to each possible outcome, that is, in our case, sales.

When a model is mis-specified, it means that the posited probability density function differs from the true probability density function. Models can be mis-specified for a variety of reasons, for example, lack of historical data to calibrate the sales response function, unreliable data collection that can lead to erroneous conclusions, or errors in the estimation procedure. To illustrate the need for robustness in the case of pricing decisions, consider the situation where a manager chooses the price  $p$  based on the maximization of the expected profit,

$$\text{Max}_p \int x(p - c)g(x|p)dx, \quad (1)$$

where  $c$  is the product's marginal cost,  $x$  unit sales, and  $g(x|p)$  is the *posited* probability function of sales. When the posited model is correctly specified, that is when

$$g(x|p) = f(x|p) \quad (\text{Correctly Specified Model}),$$

where  $f(x|p)$  is the true probability function of sales, and then the pricing strategy determined by (1) yields the highest expected profit. On the contrary, when the posited model is mis-specified, that is when

$$g(x|p) \neq f(x|p) \quad (\text{Incorrectly Specified Model}),$$

and then profit goes down because the manager should have determined the optimal pricing strategy

by maximizing

$$\max_p \int x(p - c)f(x|p)dx, \quad (2)$$

instead of maximizing (1); that is, price optimization under the mis-specified model can lead to either underpricing the product when true demand is more favorable to the firm than initially anticipated or to overpricing the product when true demand is more adversarial to the firm than initially anticipated.

The dilemma, however, is that the manager might not be able to discover  $f(x|p)$  prior to choosing the product's price, not unlike when launching a new product or entering a foreign market with limited data, or he or she must commit to a price because of a contractual agreement between channel partners before market data are available. Thus, on the one hand, the manager cannot choose price based on  $f(x|p)$ , but on the other hand, it is unclear which probability density function should be used to inform this decision under uncertainty. As a result, the robust pricing literature recommends making pricing strategies robust to distributional assumptions. To do so, and following Wald (1950), it introduces the minimax operator (or maximin in our case), such that (1) becomes

$$\max_p \min_g \int x(p - c)g(x|p)dx. \quad (3)$$

We operationalize robustness by applying  $H^\infty$  control (see, e.g., Bernhard 2002, Basar and Bernhard 2008, and Hansen and Sargent 2008).

## 2.1. $H^\infty$ Robust Control

$H^\infty$  control is a robust control framework that designs policies to limit the worst-case amplification of exogenous disturbances on a specified performance output. In a pricing context, exogenous disturbances correspond to demand shocks and performance output to profit. Specifically, given  $\gamma$  (Bernhard 2002), the  $H^\infty$  literature presents the control problem as a minimization, where the  $H^\infty$  controller  $K$  enforces the relationship

$$\inf_K \sup_w \frac{\|z\|^2}{\|u\|^2} \leq \gamma^2, \quad (4)$$

where  $\frac{\|z\|^2}{\|u\|^2}$  is the criterion used in this literature, with  $u$  as the disturbance,  $\|z\|^2$  as the performance metric, and  $\gamma$  as the level of robustness. The controller is designed to guarantee some performance against model uncertainty and unknown but energy-bounded disturbances (Doyle et al. 1989, Zhou et al. 1996). The literature on robust control (see, e.g., Basar and Bernhard 2008) has shown that this problem lends itself to a game-theoretic interpretation where the controller acts as a player selecting strategies to guard against an adversarial "Nature" choosing the worst-case disturbances,

that is,

$$\min_K \max_u \|z\|^2 - \gamma^2 \|u\|^2.$$

For a pricing problem with disturbance  $u$  and robustness level  $\gamma^2 = 1/\theta$ , the determination of the controller is characterized by

$$\max_p \min_u \left\{ (p - c)h(p, u) + \frac{u^2}{\theta} \right\}, \quad (5)$$

with  $h(p, u) = d(p) + u$ . Note that in equilibrium,  $u^*$  will be negative, as we will show, which will create a depressive force on (5).<sup>1</sup> In the next section, we demonstrate the value of this approach in the case of a vertically integrated channel.

## 3. Robustness in a Vertically Integrated Channel

### 3.1. Pricing Under the Nominal Model

Consider the situation where the manager of a vertically integrated channel does not have any concern for model uncertainty and thus does not seek robustness for the product's price. However, the product's demand is uncertain because it is perturbed by random shocks, that is,  $\epsilon_0$ , that are presumably drawn from a known distribution function. Without loss of generality, we assume that demand shocks are drawn from a normal distribution with zero mean such that demand is

$$x = d(p) + \epsilon_0, \quad (6)$$

where  $d(p)$  could be the result of market research conducted prior to making pricing decisions, with  $d'(p) \leq 0$ . Applying expected profit maximization to determine the optimal price, that is,

$$p^* = \underset{p}{\operatorname{ArgMax}} \mathbb{E}(\Pi) = \underset{p}{\operatorname{ArgMax}} d(p)(p - c), \quad (7)$$

we find that the optimal pricing strategy is

$$p^* = \frac{c \times \eta}{1 + \eta}, \quad \text{with} \quad \eta = \frac{d'(p^*) \times p^*}{d(p^*)} < 0. \quad (8)$$

This pricing strategy corresponds to the Lerner's index, which requires  $\eta < -1$ , a condition supported by empirical research. Bijmolt et al. (2005) documented, for instance, that "across a set of 1851 price elasticities based on 81 studies, the average price elasticity is  $-2.62$ ." Moreover, because all demand functions considered in this research are log-concave, the profit function  $\pi(p) = (p - c)d(p)$  is quasi-concave, ensuring that the first-order condition yields a unique profit-maximizing price.

### 3.2. Robust Pricing Strategy

We now relax the assumption that demand shocks  $\epsilon_0$  are drawn from a known distribution and consider that they are instead drawn from an alternative yet unknown distribution; for example, the manager does

not fully trust the results of the marketing research that informed the estimation of the demand function (6). Specifically, we assume that demand shocks are

$$\epsilon_0 = \epsilon + u, \quad (9)$$

where  $\epsilon$  is assumed to be distributed with a zero mean and where  $u$  is the source of model uncertainty in that it perturbs demand in a random yet unknown way, making the nominal model (6) mis-specified. Assuming that  $\epsilon \neq 0$  is not necessary for our results to hold but allows an elegant interpretation of  $u^2$ , as demonstrated by Hansen and Sargent (2008), which we provide shortly. As a result, we rewrite the nominal model (6) as

$$x = h(p, u) + \epsilon, \quad (10)$$

which implies that demand and the impact of  $p$  on it are unknown because of the disturbance  $u$ .

To introduce robustness in the pricing strategy given the sales response function (10), we follow the robust control approach and replace the firm's expected profit objective function by its robust counterpart, that is,

$$\Pi_R^{VI} = \min_u h(p, u)(p - c) + \frac{u^2}{\theta_I}, \quad (11)$$

where  $\theta_I > 0$  is the level robustness chosen by the manager for the vertically integrated channel. The mathematical foundations of (11), and in particular of  $\frac{u^2}{\theta_I}$ , can be found in Basar and Bernhard (2008), whereas the economic foundations can be found in Hansen and Sargent (2008). In particular, Hansen and Sargent (2008) showed that  $u^2/\theta_I$  informs the relative entropy that the manager envisions between the probability measure induced by the nominal model and the probability measure induced by sales' true but unknown distribution. Specifically, to connect the  $H^\infty$  control to relative entropy, one can think of two equivalent ways to bound misspecification. The  $H^\infty$  view gives nature an energy budget and leads to an inner problem of the form  $\min_u \{(p - c)h(p, u) + u^2/\theta\}$ . The relative-entropy view instead gives nature a model-distance budget around a nominal probability measure  $P_0$  such that the Kullback-Leibler (KL) divergence between the nominal model and the alternative model is bounded. Formally, the Kullback-Leibler divergence is defined by

$$D_{KL}(Q \| P_0) = \mathbb{E}_Q \left[ \log \left( \frac{dQ}{dP_0} \right) \right] \geq 0,$$

which quantifies how far an alternative model  $Q$  is from  $P_0$ . A distributionally robust inner problem with a KL ball  $\{Q : D_{KL}(Q \| P_0) \leq \varphi\}$  is equivalent to the penalized form

$$\min_Q \mathbb{E}_Q[\Pi] + \frac{1}{\kappa} D_{KL}(Q \| P_0), \quad \kappa > 0,$$

where larger  $\kappa$  means that deviations from the nominal model are penalized less (Jacobson 1973, Whittle 1990,

Hansen and Sargent 2008). Said differently, in our study, nature chooses  $u$  to depress profit but pays a quadratic cost  $u^2/\theta$ , which decreases as  $\theta$  increases.

In the case of the vertically integrated channel, setting  $\theta_I = 0^2$  means that the relative entropy between  $g(x|p)$  and  $f(x|p)$  is zero, that is,  $g(x|p) = f(x|p)$ , which implies that the manager assumes that there is no need for robustness because the nominal model correctly captures the true demand and as a result chooses the non-robust pricing strategy:

$$p^* = \frac{c\eta}{1 + \eta}.$$

In contrast,  $\theta_I > 0$  implies that the relative entropy between  $g(x|p)$  and  $f(x|p)$  differs from zero, that is,

$$g(x|p) \neq f(x|p),$$

which means that the manager seeks robustness. Although restrictions are imposed on the structure of  $u$ , the term  $u^2/\theta_I$  implies that the manager seeks a pricing strategy that is robust to all possible models that have a finite entropy with the nominal model. Specifically, a low  $\theta_I$  implies a low level of robustness and that the entropy between the considered model and the true model is low. Alternatively, a high  $\theta_I$  implies a high level of robustness and that the entropy between with the true model is large.

The optimal robust pricing strategy is characterized by

$$p^*(\theta_I) = \underset{p}{\text{ArgMax}} \Pi_R^{VI},$$

which, with (11), defines the maximin problem faced by the manager, that is,

$$\underset{p}{\text{Max}} \underset{u}{\text{Min}} h(p, u)(p - c) + \frac{u^2}{\theta_I}. \quad (12)$$

To obtain the optimal robust pricing strategy, we first solve for  $u$  and then replace  $u^*$  in (12). We then differentiate the resulting expression with respect to  $p$  to obtain the first-order condition; at the same time, we use the  $H^\infty$  criterion from robust control (see, e.g., Bernhard 2002 and Basar and Bernhard 2008) to characterize the optimal level of robustness for the vertically integrated channel, that is,

$$\frac{h(p^*, u^*)(p^* - c)}{u^{2*}} = \theta_I^{-1}, \quad (13)$$

such that the following two conditions hold true in equilibrium, that is,

$$\theta_I = \frac{4}{3} \times \frac{d(p^*)}{(p^* - c)} \text{ and } d(p^*) - (p^* - c) \left( d'(p^*) - \frac{\theta_I}{2} \right) = 0,$$

which yield the following proposition.



**Proposition 1.** Based on the nominal model  $d(p)$  and the firm's marginal cost, the level of robustness chosen by the manager in the vertically integrated channel is

$$\theta_I = -\frac{4}{3} \times \frac{(1+3\eta)}{c} \times d(p^*). \quad (14)$$

As a result, the optimal robust pricing strategy for the vertically integrated channel is

$$p^* = \frac{3 \times c \times \eta}{1+3\eta}. \quad (15)$$

Following the extant marketing literature on price elasticity, we assume that  $\eta < -1$  (see, e.g., Bijmolt et al. 2005)<sup>3</sup> and find, per our first Proposition, that the optimal robust price is lower than the optimal non-robust price because

$$\underbrace{\frac{3 \times c \times \eta}{1+3\eta}}_{\text{Robust Price}} - \underbrace{\frac{c \times \eta}{1+\eta}}_{\text{Non-Robust Price}} = \frac{2c\eta}{1+4\eta+3\eta^2} < 0.$$

By being robust, the manager adjusts the product's price in anticipation of market conditions that are more adversarial than posited by the nominal model, that is,  $d(p)$ , and thus sets a price that is lower than the non-robust price.

Furthermore, we note that introducing robustness into pricing decisions because of model uncertainty is traditionally achieved by assuming that the true model is not known but is included in a set of possible models called the *uncertainty set* (Ben-Tal and Nemirovski 1998). Both the minimax regret approach (see, e.g., Bergemann and Schlag 2011) and the robust mathematical programming approach (see, e.g., Besbes and Zeevi 2011) consider that this uncertainty set is exogenously given to the manager. Although our approach also allows for this possibility as well (which happens when  $\theta_I$  is not tuned), it also allows the decision-maker to tune the level of robustness  $\theta_I$ , which informs the entropy between the nominal and true but unknown probability distribution functions of sales (see, e.g., Hansen and Sargent 2008). As a result, by varying  $\theta_I$ , the manager considers different uncertainty sets that include all possible distribution functions that have a finite entropy with the true model. Therefore, the uncertainty set is endogenously determined.

Lastly, the condition

$$\theta_I = \frac{4}{3} \times \frac{d(p^*)}{(p^* - c)}$$

yields that the level of robustness that should be chosen by the manager is inversely proportional to the average unit margin that is anticipated by the nominal model  $d(p)$ , that is,  $\frac{p^*-c}{d(p^*)}$ . This insight yields a simple heuristic that the lower the unit margin, the higher the level of robustness. The intuition for this insight is that

unforeseen demand shocks will be more detrimental to profit when the variable cost  $c$  is high than when it is low. The manager should then seek more robustness when the variable cost is high. Based on these results, we now explore the value of robustness in the vertically integrated channel.

### 3.3. The Value of Robustness in a Vertically Integrated Channel

To characterize the value of robustness in a vertically integrated channel, we evaluate the ex post profits that are obtained with the non-robust price and robust price, respectively. Note that the firm's profit is

$$\Pi = (d(p) + \rho)(p - c), \quad (16)$$

where  $\rho$  is the ex post realization of the demand shock. When  $\rho < 0$ , it means that the realization of the demand shock was more adversarial than initially assumed by the nominal model; that is, the model was incorrectly specified at the time the price was determined. Conversely, when  $\rho > 0$ , the realization of the demand shock was more favorable than initially assumed by the nominal model; that is, the model was incorrectly specified at the time the price was determined, but in favor of the firm. Thus, the profit level attained when using the non-robust price  $p = p_{NR} = \frac{c\eta}{1+\eta}$  is

$$\Pi_{NR} = \left( d\left(\frac{c\eta}{1+\eta}\right) + \rho \right) \left( \frac{c\eta}{1+\eta} - c \right). \quad (17)$$

Similarly, the profit level attained when using the robust price  $p = p_R = \frac{3c\eta}{1+3\eta}$

$$\Pi_R = \left( d\left(\frac{3c\eta}{1+3\eta}\right) + \rho \right) \left( \frac{3c\eta}{1+3\eta} - c \right). \quad (18)$$

Furthermore, because  $p_R < p_{NR}$ , we have that  $d(p_R) > d(p_{NR})$  and obtain the following proposition.

**Proposition 2.** The robust pricing strategy yields a higher profit than the non-robust pricing strategy when  $\rho \leq \rho_I$ , where

$$\rho_I = \left( \frac{d(p_{NR})}{1+\eta} - \frac{d(p_R)}{1+3\eta} \right) \times \frac{1+4\eta+3\eta^2}{2\eta}. \quad (19)$$

To illustrate the value of the results of Proposition 1 and Proposition 2, we provide in Table 1 below the optimal robust and non-robust pricing strategies as well as the levels of robustness that should be chosen by the firm and the threshold  $\rho_I$  for three commonly used demand functions, that is, the linear demand function, the iso-elastic demand function, and the exponential demand function, that is,  $d(p) = v - \alpha \times p$ ,  $d(p) = v \times p^{-\alpha}$  and  $d(p) = ve^{-\alpha \times p}$ , respectively.

Table 1 reveals two important insights. First, we see that the levels of robustness differ, which is as it should be given results from Proposition 1. Furthermore, we

**Table 1.** Summary of Results for Different Nominal Models

|                               | Linear demand                | Iso-elastic                                                                                                                       | Exponential                                |
|-------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| $d(p)$                        | $v - \alpha \times p$        | $v \times p^{-\alpha}$                                                                                                            | $v \times e^{-\alpha \times p}$            |
| $\theta_I$                    | $4\alpha$                    | $4v\alpha \left(\frac{3c\alpha}{3\alpha-1}\right)^{-(1+\alpha)}$                                                                  | $4v\alpha e^{-(\frac{1}{\alpha}+c\alpha)}$ |
| Non-robust price ( $p_{NR}$ ) | $\frac{v+c\alpha}{2\alpha}$  | $\frac{c\alpha}{\alpha-1}$                                                                                                        | $\frac{1}{\alpha} + c$                     |
| Robust price ( $p_R$ )        | $\frac{v+3c\alpha}{4\alpha}$ | $\frac{3c\alpha}{3\alpha-1}$                                                                                                      | $\frac{1}{3\alpha} + c$                    |
| $\rho_I$                      | $-\frac{(v-c\alpha)}{4}$     | $\frac{v((1-3\alpha)(\frac{c\alpha}{\alpha-1})^{-\alpha} + 3^{-\alpha}(\alpha-1)(\frac{c\alpha}{3\alpha-1})^{-\alpha})}{2\alpha}$ | $-ve^{-v\alpha}$                           |

note that whereas in the linear demand function the level of robustness does not depend on the product's marginal cost, it will in the case of the iso-elastic and exponential demand functions. Specifically, we find that

$$\frac{\partial \theta_I}{\partial c} \leq 0,$$

that is, firms with high marginal costs should implement less robust prices.

The second insight offered by Table 1 is that in all three cases,  $\rho_I < 0$ , which means that robust pricing strategies in a vertically integrated channel produce higher profits than non-robust pricing strategies when realizations of demand shocks are more adversarial than posited by the nominal model.<sup>4</sup> This result is intuitive because seeking robustness incites the manager to lower the product's price in anticipation of sluggish market conditions. Thus, when actual market conditions are "not as bad as anticipated", that is,  $\rho > \rho_I$ , implementing a robust pricing strategy creates less value than the non-robust pricing strategy because the manager *underpriced* the product compared with what actual market conditions would have recommended. Conversely, the non-robust pricing strategy leads to *overpricing* the product when demand shocks are more adversarial (i.e.,  $\rho < \rho_I < 0$ ) than anticipated.

To summarize, when the channel is vertically integrated, (1) the manager always implements a robust pricing strategy since in equilibrium  $\theta_I > 0$ , (2) the robust pricing strategy is lower than the non-robust pricing strategy, and finally, (3) the robust pricing strategy improves profits only when market conditions are significantly worse than envisioned by the nominal model. In the next section, we analyze the equilibrium strategies and profit levels in the case of a decentralized channel involving one manufacturer and one retailer and discover new results.

#### 4. Robustness in a Decentralized Channel

The implementation of robust pricing strategies in a decentralized channel requires considering the strategic interactions between channel partners. We note that the extant robust pricing literature has focused

exclusively on situations without channel structures, thus leaving unaddressed this issue of how vertical strategic interactions would shape robust pricing strategies. In particular, how does the pursuit of robustness impact channel profits and double marginalization?

The additional complexity induced by the channel structure when choosing optimal robust pricing strategies comes from the fact that the manufacturer and the retailer might disagree on how the actual sales response model departs from the nominal model provided by market research. Formally, in a channel structure with an uncertain demand, the manufacturer chooses the wholesale price  $w$ , and the retailer chooses the retail price  $p_R$  that maximizes its respective expected profits, that is,

$$\text{Max}_w \Pi_M = \int x(w - c)g_M(x|p_R)dx, \quad (20)$$

and

$$\text{Max}_{p_R} \Pi_R = \int x(p_R - w)g_R(x|p_R)dx, \quad (21)$$

where  $g_M(\cdot)$  and  $g_R(\cdot)$  are the probability density functions that capture the possibly divergent views that the manufacturer and the retailer have regarding the true demand. Specifically, in absence of concerns for robustness, both players take the nominal model as the true data-generating process, which means that they choose their respective prices assuming that

$$g_M(\cdot) = g_R(\cdot) = g(\cdot) = f(\cdot),$$

Where, similar to the previous section,  $g(\cdot)$  is the *posited* sales' probability density function (pdf) and  $f(\cdot)$  is the *true* sales' pdf such that (20) and (21) become

$$\text{Max}_w \Pi_M = \int x(w - c)g(x|p_R)dx, \quad (22)$$

and

$$\text{Max}_{p_R} \Pi_R = \int x(p_R - w)g(x|p_R)dx, \quad (23)$$

respectively. Therefore, relaxing the assumptions that  $g_M(\cdot) = f(\cdot)$  and  $g_R(\cdot) = f(\cdot)$  implies (i) that both players might doubt that the nominal model is the true data-generating process (i.e.,  $g_M(\cdot) \neq f(\cdot)$  and  $g_R(\cdot) \neq f(\cdot)$ )

and (ii) that they might disagree on what should be the alternative probability density functions that they should consider (i.e.,  $g_M(\cdot) \neq g_R(\cdot)$ ).

Similar to the vertically integrated channel, we consider that the uncertain demand function is

$$x = d(p) + \epsilon_0, \quad (24)$$

where  $d(p)$  is the nominal demand model that could be provided by market research, with  $d'(p) \leq 0$ , and where  $\epsilon_0$  captured the random shocks that make demand uncertain, which are usually assumed to be drawn from a known distribution function. In the case of vertically integrated channels, we assumed that

$$\epsilon_0 = \epsilon + u, \quad (25)$$

where  $u(\cdot)$  was the source of model uncertainty in that it made the nominal model mis-specified. Considering model uncertainty in the decentralized channel requires allowing for the fact that the manufacturer and the retailer might have divergent views about model uncertainty in that the manufacturer might consider the demand function

$$x = h_M(p, w, u_M) = d(p) + u_M + \epsilon, \quad (26)$$

whereas the retailer might consider the demand function

$$x = h_R(p, w, u_R) = d(p) + u_R + \epsilon, \quad (27)$$

where  $u_M$  is the source of model uncertainty that perturbs demand in a random yet unknown way, according to the manufacturer, and  $u_R$  is the source of model uncertainty according to the retailer. Consequently, three scenarios are possible. In the first scenario, neither the manufacturer nor the retailer have concerns for robustness, that is,

$$u_M = u_R = 0,$$

and the optimal pricing strategies are obtained by standard expected profit maximization. Specifically, when neither the retailer nor the manufacturer seek robustness, the optimal pricing strategies are

$$w^* = \frac{c\eta}{1+\eta} \times \frac{1+\eta+\tau}{2+\eta+\tau} \quad \text{and} \quad p^* = \frac{w^*\eta}{1+\eta}, \quad (28)$$

where  $\eta = \frac{d'(p(w)) \times p(w)}{d(p(w))}$  and  $\tau = \frac{\partial \eta}{\partial w} \times \frac{w}{\eta}$  (evaluated at the optimum), which is similar to the Lerner's pricing rule in a channel setting, where  $\tau$  captures the compound effect of the wholesale price on demand through demand elasticity.

In the second scenario, only one channel partner implements a robust pricing strategy, whereas the other does not, that is,  $u_M = 0$  and  $u_R > 0$ , in which case, the retailer seeks robustness, whereas the manufacturer does not, or  $u_M > 0$  and  $u_R = 0$ , in which case, the retailer does not seek robustness, whereas the manufacturer does. Finally, in the third scenario, both

channel partners have concerns from robustness such that  $u_M > 0$  and  $u_R > 0$ , whereby either

$$u_M = u_R,$$

in which case, both the manufacturer and the retailer seek robustness and do not disagree on how these unexpected demand shocks distort demand, or

$$u_M \neq u_R.$$

in which case, both the manufacturer and the retailer seek robustness but disagree with respect to the demand function to consider for price optimization. To determine the equilibrium scenario, we derive the optimal robust pricing strategies for the manufacturer and the retailer next.

#### 4.1. Robust Pricing Strategies

Channel partners derive their respective pricing strategies through the following *maximin* optimization problems, that is,

$$w^* = \underset{w}{\text{ArgMax}} \quad \Pi_M \quad \text{and} \quad p^* = \underset{p}{\text{ArgMax}} \quad \Pi_R,$$

with

$$\Pi_M = \underset{u_M}{\text{Min}} h_M(p, w, u_M)(w - c) + \frac{u_M^2}{\theta_M}, \quad (29)$$

and

$$\Pi_R = \underset{u_R}{\text{Min}} h_R(p, w, u_R)(p_R - w) + \frac{u_R^2}{\theta_R}, \quad (30)$$

with  $h_M(\cdot)$  and  $h_R(\cdot)$  as given by (26) and (27), respectively, where the manufacturer (retailer) acts as a Stackelberg leader (follower) and determines the robust wholesale (retail) price. Hence, the manufacturer first announces the wholesale price, which is then used by the retailer to determine the retail price. We present the equilibrium strategies in the next proposition.

**Proposition 3.** *Based on the nominal model  $d(p)$ , the levels of robustness chosen by the manufacturer and the retailer are*

$$\theta_M = \frac{4d(p^*)}{3(w^* - c)} \quad \text{and} \quad \theta_R = \frac{4d(p^*)}{3(p^* - w^*)}, \quad (31)$$

respectively, and the optimal robust wholesale and retail prices are

$$w^* = \frac{3c\eta}{1+3\eta} \times \frac{1+3\eta+\gamma}{2+3\eta+\gamma} \quad \text{and} \quad p^* = \frac{3\eta w^*}{1+3\eta}, \quad (32)$$

respectively, where  $\tau = \frac{\partial \eta}{\partial w} \times \frac{w}{\eta}$ .

Proposition 3 reveals that in equilibrium, both the retailer and the manufacturer pursue robustness because  $\theta_R > 0$  and  $\theta_M > 0$ . Specifically, we find that

$$\frac{\theta_R}{\theta_M} = \frac{w^* - c}{p^* - w^*},$$

which means that the levels of robustness for the two channel partners depend on the respective unit margins

they expect to extract. Specifically, the retailer pursues more robustness than the manufacturer, that is,  $\theta_R > \theta_M$ , when  $w^* - c > p^* - w^*$  and  $\theta_R < \theta_M$  otherwise. Therefore, who seeks “more” robustness depends on how strategic interactions in the channel shape the unit margins that channel partners aim at capturing.

To illustrate the implications of Proposition 3, we compute the optimal robust and non-robust pricing strategies for the linear demand function, the iso-elastic demand function, and the exponential demand function, as well as the levels of robustness chosen by the channel partners, and report the results in Table 2. We first learn that the levels of robustness chosen by the manufacturer and the retailer can be equal or different, depending on the nature of the nominal demand function. Specifically, we find that  $\theta_M = \theta_R$  in the case of the exponential demand function, whereas  $\theta_M \neq \theta_R$  for the linear and Iso-Elastic demand functions. Furthermore, who is “more” robust also depends on the nature of the demand function, that is, the retailer in the case of the linear demand function and the manufacturer in the case of the iso-elastic demand function.

The second insight provided by Table 2 is that the robust pricing strategies in marketing channels are lower than the non-robust pricing strategies. For instance, in the case of the linear demand function, the difference between the robust and non-robust wholesale prices is

$$\underbrace{\frac{v+3c\alpha}{4\alpha}}_{\text{Robust Wholesale Price}} - \underbrace{\frac{v+c\alpha}{2\alpha}}_{\text{Non Robust Wholesale Price}} = -\frac{v-c\alpha}{4\alpha} < 0,$$

whereas the difference between the robust and non-robust retail prices is

$$\underbrace{\frac{7v+9c\alpha}{16\alpha}}_{\text{Robust Retail Price}} - \underbrace{\frac{3v+c\alpha}{4\alpha}}_{\text{Non Robust Retail Price}} = -\frac{5(v-c\alpha)}{16\alpha},$$

which is negative assuming that  $v - c\alpha > 0$ . In the case of the iso-elastic demand function, the differences

**Table 2.** Robust vs. Non-robust Pricing Strategies in Marketing Channels

| $d(p)$                | Linear demand<br>$v - \alpha \times p$ | Iso-elastic<br>$v \times p^{-\alpha}$                                                        | Exponential<br>$v \times e^{-\alpha \times p}$ |
|-----------------------|----------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------|
| Non-robust strategies |                                        |                                                                                              |                                                |
| Wholesale price       | $\frac{v+c\alpha}{2\alpha}$            | $\frac{c\alpha}{\alpha-1}$                                                                   | $\frac{1+c\alpha}{\alpha}$                     |
| Retail price          | $\frac{3v+c\alpha}{4\alpha}$           | $\frac{c\alpha^2}{(\alpha-1)^2}$                                                             | $c + \frac{2}{\alpha}$                         |
| Robust strategies     |                                        |                                                                                              |                                                |
| $\theta_M$            | $3\alpha$                              | $\frac{4v(3\alpha-1)\left(\frac{c\alpha^2}{9(1-3\alpha)^2}\right)^{-\alpha}}{c^{3+2\alpha}}$ | $4v\alpha e^{-(c\alpha+\frac{2}{3})}$          |
| $\theta_R$            | $4\alpha$                              | $4v\alpha\left(\frac{c\alpha^2}{9(1-3\alpha)^2}\right)^{-(1+\alpha)}$                        | $4v\alpha e^{-(c\alpha+\frac{2}{3})}$          |
| Wholesale price       | $\frac{v+3c\alpha}{4\alpha}$           | $\frac{3c\alpha}{3\alpha-1}$                                                                 | $\frac{1}{3\alpha} + c$                        |
| Retail price          | $\frac{7v+9c\alpha}{16\alpha}$         | $\frac{9c\alpha^2}{(1-3\alpha)^2}$                                                           | $\frac{2}{3\alpha} + c$                        |

between the robust and non-robust wholesale prices and between the robust and non-robust retail prices are

$$-\frac{2c}{1-4\alpha+3\alpha^2} \quad \text{and} \quad \frac{4c\alpha^2(2-3\alpha)}{(1-4\alpha+3\alpha^2)^2}, \quad (33)$$

respectively, which are negative for reasonable values of  $\alpha$ , that is,  $\eta = -\alpha < -1$ . Finally, in the case of the exponential demand function, the differences between the robust and non-robust wholesale prices and between the robust and non-robust retail prices, are

$$-\frac{2}{3\alpha} < 0 \quad \text{and} \quad -\frac{4}{3\alpha} < 0, \quad (34)$$

respectively. Thus, Proposition 3 generalizes to a channel setting the results of Proposition 1 in that robust wholesale and retail prices are lower than the non-robust counterparts. Most importantly, Proposition 3 provides the necessary equations to characterize the optimal robust pricing strategies in marketing channels, a result that is absent from both the robust pricing literature and the marketing channel literature. Based on these results, we now study the value of robustness in a decentralized channel.

#### 4.2. The Value of Robustness in a Decentralized Channel

To characterize the value of robustness in a decentralized channel, we evaluate the ex post profits that are obtained with the non-robust pricing strategies and robust pricing strategies, respectively. Similar to the analysis of the value of robustness in the centralized channel, we compute the manufacturer’s and retailer’s ex post profits as

$$\Pi_M = (d(p) + \rho)(w - c) \quad \text{and} \quad \Pi_R = (d(p) + \rho)(p - c), \quad (35)$$

where  $\rho$  is the realization of the demand shock. To obtain the manufacturer’s and retailer’s ex post profits under the robust pricing strategies ( $\Pi_M^R$  and  $\Pi_R^R$ ), we replace  $w$  and  $p$  with  $w_R^* = \frac{3c\eta}{1+3\eta} \times \frac{1+3\eta+\gamma}{2+3\eta+\gamma}$  and  $p_R^* = \frac{3w^*}{1+3\eta}$ , respectively, in (35). Similarly, to obtain the manufacturer’s and retailer’s ex post profits under the “standard” non-robust pricing strategies ( $\Pi_M^N$  and  $\Pi_R^N$ ), we replace  $w$  and  $p$  with  $w_N^* = \frac{c\eta}{1+\eta} \times \frac{1+\eta+\gamma}{2+\eta+\gamma}$  and  $p_N^* = \frac{w^*\eta}{1+\eta}$ , respectively, in (35). Doing so allows us to compute the difference in profits for the manufacturer and the retailer as  $\Delta_M = \Pi_M^R - \Pi_M^N$  and  $\Delta_R = \Pi_R^R - \Pi_R^N$  such that

$$\Delta_M = -\rho(w_N^* - w_R^*) + d(p_R)(w_R^* - c) - d(p_N)(w_N^* - c) \quad (36)$$

and

$$\Delta_R = \rho(p_R^* - w_R^* - (p_N^* - w_N^*)) + d(p_R)(p_R^* - w_R^*) - d(p_N)(p_N^* - w_N^*), \quad (37)$$

As a result, we obtain the following proposition for the manufacturer.



**Proposition 4.** Robust pricing strategies increase the manufacturer's profit when the realization of the demand shock is such that  $\rho < \rho_M$ , with

$$\rho_M = \frac{d(p_R)(w_R^* - c) - d(p_N)(w_N^* - c)}{w_N^* - w_R^*}. \quad (38)$$

Conversely, we obtain the following proposition regarding the retailer.

**Proposition 5.** Robust pricing strategies increase the retailer's profit when the realization of the demand shock is such that  $\rho < \rho_R$ , with

$$\rho_R = \frac{d(p_R)(p_R^* - w_R^*) - d(p_N)(p_N^* - w_N^*)}{(p_N^* - w_N^*) - (p_R^* - w_R^*)}. \quad (39)$$

The main insight revealed by (38) and (39) is that  $\rho_M \neq \rho_R$ ; that is, it is possible that being robust benefits, ex post, one channel partner but not the other; although ex ante, it is desirable for both the manufacturer and the retailer to set  $\theta_M > 0$  and  $\theta_R > 0$ , as demonstrated by Proposition 3.

To further explore the value of robustness in marketing channels, we provide in Table 3 below the threshold values  $\rho_M$  and  $\rho_R$  for the following nominal models, that is, linear, iso-elastic, and exponential.

Focusing on the linear and exponential demands, Table 3 allows us to make the following observations. We first note that although the thresholds are different,  $\rho_M$  and  $\rho_R$  in the case of the linear demand function, we find that they are the same in the case of the exponential demand function. Most interestingly, we recall from Table 1 that in the case of the linear demand function, being robust would create more ex post profit than being non-robust in the case of the vertically integrated channel when  $\rho < -\frac{(v-c\alpha)}{4}$  and  $\rho < -ve^{-v\alpha}$  in the case of the exponential demand function. Thus, two observations follow. First, in the case of the exponential demand, we observe that the threshold value such that robust pricing strategies would yield higher profits than the non-robust pricing strategies remains unchanged. Second, however, we observe that in the case of the linear demand, not only are the manufacturer's and retailer's threshold values different from the one in the case of the vertically integrated channel, but also these values are positive.

This last finding is surprising in the sense that it suggests that robust pricing strategies in the decentralized

channel can create value not only when demand is actually more adversarial than anticipated by the nominal model, as it should be, but also when demand shocks are beneficial to the firm. This was not the case in the vertically integrated channel. The rationale for this finding is that by being robust, both channel partners choose prices that are lower than the non-robust pricing strategies, which we summarize in the proposition below.

**Proposition 6.** Robustness can act as a coordinating device in marketing channels.

In channels, this downward pressure on prices creates value even when demand shocks are more beneficial to the firm because they partially mitigate double marginalization. This finding also holds true for some parameter values in the case of the iso-elastic demand. Therefore, an important insight is that in marketing channels, robust pricing strategies can create value not only by helping manage unforeseeable demand shocks but also by mitigating double marginalization.

In addition to the results provided by Table 3, we report in Table 4 the total channel profits under the robust and non-robust strategies, that is,

$$\Pi_D^R = \Pi_M^R + \Pi_R^R, \quad (\text{Channel profit of the robust decentralized channel})$$

$$\Pi_D^N = \Pi_M^N + \Pi_R^N, \quad (\text{Channel profit of the non-robust decentralized channel}),$$

respectively, as well as the ex post profits in the case of the vertically integrated channel for both robust and non-robust pricing strategies, that is,

$$\Pi_{VI}^R \quad \text{and} \quad \Pi_{VI}^N,$$

respectively.

Table 4 first confirms that robust pricing strategies can create value by mitigating double marginalization. For instance, in the case of the linear demand function, we obtain that the robust decentralized channel yields more profit ( $\Pi_D^R$ ) than the non-robust decentralized channel ( $\Pi_D^N$ ) when realizations of demand shocks are lower than  $\frac{3}{16}(v - c\alpha)$ , that is,

$$\Pi_D^R > \Pi_D^N \quad \text{when} \quad \rho < \frac{3}{16}(v - c\alpha),$$

which means that robust pricing strategies in the decentralized channel can create more value than the non-robust pricing strategies not only when demand shocks are adversarial but also when they are favorable to the firm (as long as  $\rho < \frac{3}{16}(v - c\alpha)$ ), because robust pricing strategies partially mitigate double marginalization compared with the non-robust case.

More surprisingly, we find that under certain circumstances, robust pricing strategies can lead to a situation where the total channel profit in the robust

**Table 3.** Threshold Values for the Decentralized Channel

|          | Linear demand              | Iso-elastic                                                                                                                                                                                | Exponential      |
|----------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| $\rho_M$ | $\frac{(v-c\alpha)}{16}$   | $\frac{v \left( 9^{-\alpha} (\alpha-1) \left( \frac{a^2 c}{(1-3\alpha)^2} \right)^{-\alpha} + (1-3\alpha) \left( \frac{a^2 c}{(a-1)^2} \right)^{-\alpha} \right)}{2\alpha}$                | $-ve^{-v\alpha}$ |
| $\rho_R$ | $\frac{11(v-c\alpha)}{16}$ | $\frac{v \left( \left( \frac{a^2 c}{(a-1)^2} \right)^{-\alpha} - 3^{1-2\alpha} \left( \frac{a^2 c}{(1-3\alpha)^2} \right)^{-\alpha} \right)}{\frac{3}{(1-3\alpha)^2} - \frac{1}{(a-1)^2}}$ | $-ve^{-v\alpha}$ |

**Table 4.** Ex Post Channel Profits Under Robust and Non-robust Strategies

|                             | Linear demand                                       | Iso-elastic                                                                                                                                                                               | Exponential                            |
|-----------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Decentralized channels      |                                                     |                                                                                                                                                                                           |                                        |
| Robust ( $\Pi_D^R$ )        | $\frac{7(v-\alpha)(9(v-\alpha)+16\rho)}{256\alpha}$ | $\frac{3(6\alpha-1)\left(v+9^{\frac{1}{\alpha}}\rho\left(\frac{\alpha^2}{(1-\alpha)^2}\right)^{\frac{1}{\alpha}}\right)\left(\frac{\alpha^2}{(1-3\alpha)^2}\right)^{1-\alpha}}{\alpha^2}$ | $\frac{2(e^{-v\alpha}+\rho)}{3\alpha}$ |
| Non-robust ( $\Pi_D^N$ )    | $\frac{3(v-\alpha)(v-\alpha+4\rho)}{16\alpha}$      | $\frac{(2\alpha-1)\left(v+\rho\left(\frac{\alpha^2}{(1-\alpha)^2}\right)^{\frac{1}{\alpha}}\right)\left(\frac{\alpha^2}{(1-\alpha)^2}\right)^{1-\alpha}}{\alpha^2}$                       | $\frac{2(e^{-v\alpha}+\rho)}{\alpha}$  |
| Centralized channels        |                                                     |                                                                                                                                                                                           |                                        |
| Robust ( $\Pi_{VI}^R$ )     | $\frac{(v-\alpha)(3(v-\alpha)+4\rho)}{16\alpha}$    | $\frac{v3^{-\alpha}\left(\frac{\alpha}{3\alpha-1}\right)^{1-\alpha}+\rho}{3\alpha-1}c$                                                                                                    | $\frac{ve^{-v\alpha}+\rho}{3\alpha}$   |
| Non-robust ( $\Pi_{VI}^N$ ) | $\frac{(v-\alpha)(v-\alpha+2\rho)}{4\alpha}$        | $\frac{v\left(\frac{\alpha}{\alpha-1}\right)^{1-\alpha}+\rho}{\alpha-1}c$                                                                                                                 | $\frac{ve^{-v\alpha}+\rho}{\alpha}$    |

decentralized structure exceeds the total profit attained in the robust vertically integrated channel, which we summarize in the next proposition.

**Proposition 7.** *Robust pricing strategies can mitigate double marginalization, that is,*

$$\Pi_D^R > \Pi_{VI}^R, \quad (40)$$

when  $\rho > \tilde{\rho}$

The threshold value  $\tilde{\rho}$  depends on the demand function. For instance, assuming that  $v - \alpha > 0$ , we find that in the linear demand function,

$$\Pi_D^R > \Pi_{VI}^R, \quad \text{when } \rho > -\frac{5}{16}(v - \alpha), \quad (41)$$

whereas in the case of the iso-elastic demand function,

$$\Pi_D^R > \Pi_{VI}^R, \quad \text{when } \rho > \frac{3^{-2\alpha-1}\left(\left(\frac{\alpha^2 c}{(1-3\alpha)^2}\right)^{-\alpha}(v - 6\alpha v) + (3\alpha - 1)v\left(\frac{\alpha c}{9\alpha-3}\right)^{-\alpha}\right)}{\alpha}, \quad (42)$$

and in the case of the exponential demand function,

$$\Pi_D^R > \Pi_{VI}^R, \quad \text{when } \rho > -ve^{-v\alpha}. \quad (43)$$

Therefore, the robust decentralized channel structure can generate more profit than the robust centralized channel structure when firms implement robust pricing strategies and when demand shocks happen to be above a certain threshold.

This last observation contrasts with the extant literature on marketing channel coordination, which despite few notable exceptions (see, e.g., Desai et al. 2004) finds that the decentralized structure is less efficient (lower profits) than the centralized structure in absence of specific contracts or coordination mechanisms because of double marginalization. The rationale for our finding as for why the robust decentralized channel can yield

higher profit than the robust centralized channel comes from the nature of demand shocks. Specifically, when realizations of demand shocks are “not as bad” as anticipated, for example,  $\rho > -\frac{5}{16}(v - \alpha)$ , for instance, as in the case of the linear demand function, it means that market conditions were relatively favorable and that the manager of the vertically integrated channel could have chosen a higher price to maximize profit but did not. Because of channel decentralization (and hence, double marginalization), the robust retail price is higher than the robust retail price obtained in the vertically integrated channel. As a result, the robust decentralized channel yields a higher profit because the vertically integrated channel structure makes the manager *underprice* the product. Say, differently, that the optimal retail price that should have been chosen under perfect knowledge of demand should have been higher than the robust pricing strategy implemented under the vertically integrated channel. In this situation, the robust decentralized channel is thus better because of double marginalization.

Conversely, when realizations of demand shocks are “bad,” for example,  $\rho < -\frac{5}{16}(v - \alpha)$  in the case of the linear demand function, then the robust vertically integrated channel dominates the robust decentralized channel because of double marginalization.

**4.2.1. Numerical Example.** We close this section by providing a numerical example of our results. Assume that  $v = \$1,000$ ,  $\alpha = 2$ , and  $c = \$400$ . From Table 5, we obtain the optimal robust and non-robust pricing strategies in the vertically integrated channel.

From Table 6, we obtain the robust and non-robust pricing strategies in the decentralized marketing channel.

**Table 6.** Robust vs. Non-robust Pricing Strategies in Marketing Channels: A Numerical Example

|                       | Linear demand | Iso-elastic | Exponential |
|-----------------------|---------------|-------------|-------------|
| Non-robust strategies |               |             |             |
| Wholesale price       | \$450         | \$800       | \$400.5     |
| Retail price          | \$475         | \$1,600     | \$401       |
| Robust strategies     |               |             |             |
| Wholesale price       | \$425         | \$480       | \$400.2     |
| Retail price          | \$443         | \$576       | \$400.3     |

**Table 5.** Optimal Prices in the Vertically Integrated Channel

|                               | Linear demand | Iso-elastic | Exponential |
|-------------------------------|---------------|-------------|-------------|
| Non-robust price ( $p_{NR}$ ) | \$450         | \$800       | \$400.5     |
| Robust price ( $p_R$ )        | \$425         | \$480       | \$400.1     |

## 5. Conclusion

Imagine that you are a brand manager responsible for selling a radically new technology or that you are contemplating entering a new foreign market. Results from market research provide you with some information about the possible demand function for your product. However, you know that even though some models are useful, all models are wrong (Box 1976). Thus, you know that in all likelihood, the market research information that you have access to, that is, the only information you can use to make pricing decisions, is most likely incorrect, but you are unsure how incorrect it is. What should be the price that you will offer the retailer or distributor responsible for selling your product? Similarly, as the retailer, what should be the retail price that you should offer customers?

The extant robust pricing and marketing channel literatures do not provide any guideline to address these managerial questions. On the one hand, the robust pricing literature does not inform how strategic interactions in a channel should impact robust pricing strategies; for example, should the manufacturer be more or less robust than the retailer? On the other hand, even though the marketing channel literature informs how to account for strategic interactions between channel partners, it remains silent on how manufacturers and retailers should make their pricing decisions robust to model uncertainty. To bridge this gap between the two literatures and provide managerial guidelines to implement robust pricing strategies in marketing channels, we propose a novel model that allows channel partners to strategically price products under model uncertainty, where a model is understood as a probability density function over possible sales outcomes.

To this end, we introduce the parameter ( $\theta$ ), which captures a firm's tolerance for model uncertainty and can be interpreted as a measure of ambiguity aversion. Conceptually,  $\theta$  plays a role analogous to the parameter governing multiplier preferences in Hansen and Sargent (2008) (see also Strzalecki (2011) for an axiomatic foundation). In managerial terms, higher robustness reflects greater caution or ambiguity aversion in decision-making. Specifically, managers place less trust in a specific demand model and instead guard against misspecification. Such a posture may be adopted, for instance, in environments where demand forecasts are volatile, historical data are limited or noisy, or market conditions are undergoing structural change (e.g., new products, new channels, or emerging customer segments). Therefore, differences in  $\theta$  across channel partners may capture heterogeneity in data quality, analytical sophistication, or access to customer information. For example, a manufacturer with access to rich customer data and advanced analytics may operate with a

lower robustness parameter, whereas a retailer facing fragmented demand signals or limited visibility into end-customer behavior may optimally adopt a higher degree of robustness.

Building on this framework, our analysis provides three key managerial insights. First, we show that the levels of robustness chosen by each channel partner depend on the respective average unit margins they expect to extract. Specifically, the player who should be the most robust is the one that expects to capture a lower unit margin. This result is intuitive because the channel partner with lowest unit margin is the one who has the most to lose from unforeseen demand shocks that are due to mis-specified models. Second, we show that robust pricing strategies in marketing channels are lower than non-robust pricing strategies, which generalizes to a channel setting an important result from the robust pricing literature. Finally, and most importantly, we show that the value of robustness in marketing channels is twofold. On the one hand, and similar to the value of robustness in vertically integrated channels, we find that the first driver of value of robustness in marketing channels is the ability to manage unforeseeable demand shocks. On the other hand, however, we also find that in addition to this first driver, being robust in marketing channel can under certain circumstances mitigate double marginalization. Thus, although robustness can have negative effects on a monopolist's profit, especially when demand conditions are better than anticipated, we find that being robust in channels can improve profits of both the manufacturer and the retailer.

Our analysis opens several avenues for future research. First, although our model assumes that both manufacturer and retailer are equally uncertain about the demand distribution, in practice one party may have more information because of differences in data access or proximity to consumers. For example, large retailers often collect point-of-sale data that could make them better positioned to assess demand risks, whereas manufacturers may rely on proprietary market studies or global trend data (Singh et al. 2024). Exploring robust pricing strategies under asymmetric information, where one party possesses richer or noisier demand signals than the other, represents a promising avenue for future research. Such an extension would allow for studying not only how robustness interacts with double marginalization but also how information-sharing agreements and incentive contracts might be designed to mitigate inefficiencies in decentralized channels. Second, a natural extension would be to generalize the framework to multiple retailers and/or multiple manufacturers. Real-world channels are rarely dyadic, and competitive dynamics among retailers or upstream suppliers may significantly alter

the incentives to adopt robust pricing. Finally, it would be fruitful to incorporate learning about the demand distribution into the robust pricing problem. In many industries, channel partners collect sales data and update their beliefs over time. Studying a dynamic robust pricing model in which agents gradually learn about the underlying demand process and possibly behave strategically in how they share or withhold information would bring the framework closer to practice.

## Appendix

### Proof Proposition 1

The maximin problem faced by the manager is

$$\max_p \min_u (d(p) + u)(p - c) + \frac{u^2}{\theta_I}. \quad (\text{A.1})$$

Differentiating (A.1) with respect to  $u$  and equating the resulting expression to zero yields that

$$(p - c) + 2\frac{u}{\theta_I} = 0 \implies u^* = -\frac{p - c}{2}\theta_I,$$

which we replace in (A.1) to obtain

$$\max_p \left( d(p) + \left( -\frac{p - c}{2}\theta_I \right) \right) (p - c) + \frac{\left( -\frac{p - c}{2}\theta_I \right)^2}{\theta_I}. \quad (\text{A.2})$$

Differentiating (A.2) with respect to  $p$  yields that at the optimum,

$$d(p^*) - (p^* - c) \left( d'(p^*) - \frac{\theta_I}{2} \right) = 0. \quad (\text{A.3})$$

Furthermore, the  $H^\infty$  criterion at the optimum provides the additional condition

$$\begin{aligned} \frac{(d(p^*) + u^*)(p^* - c)}{u^{2*}} &= \frac{1}{\theta_I} \\ \frac{2 \left( \frac{d(p^*)}{p^* - c} - \theta_I \right)}{\theta_I^2} &= \frac{1}{\theta_I} \\ \implies \theta_I &= \frac{4d(p^*)}{3(p - c)}. \end{aligned}$$

Replacing  $\theta_I = \frac{4d(p^*)}{3(p - c)}$  into (A.3) yields

$$\begin{aligned} d(p^*) - (p^* - c) \left( d'(p^*) - \frac{\frac{4d(p^*)}{3(p - c)}}{2} \right) &= 0 \\ \implies \frac{d(p^*)}{3} - (p^* - c)d'(p^*) &= 0 \\ \implies \frac{d(p^*)}{3} \frac{p^*}{d(p^*)} - (p^* - c)d'(p^*) \frac{p^*}{d(p^*)} &= 0 \\ \implies \frac{p^*}{3} - (p^* - c)\eta &= 0 \\ \implies p^* &= \frac{3c\eta}{1 + 3\eta}, \end{aligned}$$

where  $\eta = \frac{\partial d(p^*)}{\partial p^*} \times \frac{p^*}{d(p^*)}$ . Replacing  $p^* = \frac{3c\eta}{1 + 3\eta}$  into  $\theta_I = \frac{4d(p^*)}{3(p - c)}$  yields that

$$\begin{aligned} \theta_I &= \frac{4d(p^*)}{3(p^* - c)} \\ \implies \theta_I &= \frac{4d(p^*)}{3 \left( \frac{3c\eta}{1 + 3\eta} - c \right)} \\ \implies \theta_I &= -\frac{4}{3} \times \frac{1 + 3\eta}{c} d(p^*). \end{aligned}$$

### Proof Proposition 2

This is straightforward by computing the difference between (17) and (18).

### Proof Proposition 3

We solve the game backward. The retailer's robust objective function is

$$\max_p \min_{u_R} (d(p) + u_R(w, p_R))(p - w) + \frac{u_R(w, p_R)^2}{\theta_R}. \quad (\text{A.4})$$

We differentiate (A.4) with respect to  $u_R$  and equate the resulting expression to zero to obtain that  $u_R^* = -\frac{p - w}{2}\theta_R$ , which we replace in (A.4). We then differentiate the resulting expression with respect to  $p$  to obtain the first order condition:

$$d(p^*(w)) + (p^*(w) - w) \left( d'(p^*(w)) - \frac{\theta_R}{2} \right) = 0. \quad (\text{A.5})$$

Furthermore, the  $H^\infty$  criterion for the retailer provides the additional condition

$$\begin{aligned} \frac{h(p^*(w), u_R^*(p^*(w)))(p^*(w) - w)}{u_R^{2*}(p^*(w))} &= \theta_R^{-1} \\ \implies -2 \left( \theta_R + \frac{2d(p^*(w))}{p^*(w) - w} \right) &= \theta_R \end{aligned} \quad (\text{A.6})$$

We then solve for  $p^*$  and  $\theta_R$  that satisfy the system of Equations (A.5) and (A.6), which give

$$\theta_R = -\frac{4}{3} \times \frac{(1 + 3\eta)}{w} \times d(p^*(w)) \quad \text{and} \quad p^*(w) = \frac{3 \times w \times \eta}{1 + 3\eta},$$

$$\text{where } \eta = \frac{\partial d(p)}{\partial p} \times \frac{p}{d(p)},$$

where  $\eta$  is evaluated at  $p^*(w)$ . Next, we replace the retailer's reaction function in the manufacturer's robust objective function such that

$$\max_w \min_{u_M} (d(p^*(w)) + u_M(w, p^*(w)))(w - c) + \frac{u_M(w, p^*(w))^2}{\theta_M}. \quad (\text{A.7})$$

We then differentiate (A.7) with respect to  $u_M$  and equate the resulting expression to zero to obtain that  $u_M^* = -\frac{w - c}{2}\theta_M$ , which we replace in (A.7). We then differentiate the resulting expression with respect to  $w$  to obtain the first-order condition:

$$d(p^*(w)) + (w - c) \left( d'(p^*(w)) \left( p^*(w) + \frac{3w \frac{\partial \eta}{\partial w}}{(1 + 3\eta(w^2))} \right) - \frac{\theta_M}{2} \right) = 0. \quad (\text{A.8})$$



Furthermore, the  $H^\infty$  criterion for the manufacturer provides the additional condition:

$$\theta_M = \frac{4d\left(\frac{3 \times w \times \eta}{1+3\eta}\right)}{3(w-c)} \quad (\text{A.9})$$

We then solve for  $w^*$  and  $\theta_M$ , which satisfy the system of Equations (A.8) and (A.9), which give

$$\theta_M = \frac{4d(p^*)}{3(w^* - c)} \quad \text{and} \quad \theta_R = \frac{4d(p^*)}{3(p^* - w^*)}, \quad (\text{A.10})$$

respectively, where

$$w^* = \frac{3c\eta}{1+3\eta} \times \frac{1+3\eta+\gamma}{2+3\eta+\gamma} \quad \text{and} \quad p^* = \frac{3\eta w^*}{1+3\eta}, \quad (\text{A.11})$$

respectively, with  $\gamma = \frac{\partial \eta}{\partial w} \times \frac{w}{\eta}$ .

## Endnotes

<sup>1</sup> Another example of such a formulation can be found in Roseta-Palma and Xepapadeas (2004) in the case of water management.

<sup>2</sup> Note that in this case, the term  $\frac{u^2}{\theta} \rightarrow +\infty$ , which means that the “minimizing agent” cannot achieve the objective of minimizing the inner problem  $\min_u \{(p-c)h(p,u) + u^2/\theta\}$ .

<sup>3</sup> Bijmolt et al. (2005) reported that “the average price elasticity is  $-2.62$ .”

<sup>4</sup> The condition  $\rho_l < 0$  was explored numerically for the iso-elastic demand function.

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